ECSE 331

PSPICE Assignment No. 4

Design of a BJT Amplifier in PSPICE[1]

Objective

Design a BJT amplifier with specific requirements. Validate the design using theoretical calculations and PSPICE simulations. This assignment is much more free-flowing than previous ones, as the student is expected to solve a design problem using his/her own unique design method.

Write-Up Requirements

A good laboratory report should contain a brief description of what the experiment was about, including circuit diagrams, and what you did, your data, your results, and anything else called for in the assignment, such as questions inserted in the laboratory. Answers to these questions require observations that need to be made at the time you do the experiment. The laboratory report should be written using the IEEE paper style consisting of a double-column single-space format, and must adhere to the following:

1. Title page - Title of the assignment/project, authors' name, and course name.
2. Abstract - Abstract of the assignment/project report.
3. Introduction
4. Main body of the assignment/project report including figures.
6. Conclusions
7. References
8. Appendices

Task

The task is to design a BJT amplifier circuit that must be capable of operating off a 9 V battery. The magnitude of the voltage gain should be 50 V/V with a 10% gain tolerance while driving a load of 10 kΩ. In other words, the magnitude of the gain can be anywhere between 45 V/V and 55 V/V.

In addition, your amplifier must be capable of performing amplification over a 10 kHz bandwidth (i.e. 3-dB bandwidth has to be greater than 10 kHz) and have an input resistance of at least 1 kΩ .

The design of the circuit, choice of circuit components and their values should first be validated using theoretical analysis of the circuit and then using simulation results.

Investigate the temperature characteristics of your amplifier. That is, demonstrate its performance variations at different operating temperatures and explain the results.

This concludes this assignment.

[1] The choice of PSPICE as the simulation software is not strict. The student may use any other simulation tool he/she is comfortable with.